



Google input tools

Google input tools allows for transliteration, handwriting and virtual keyboards for a many languages <https://digit.in/sep19-41>

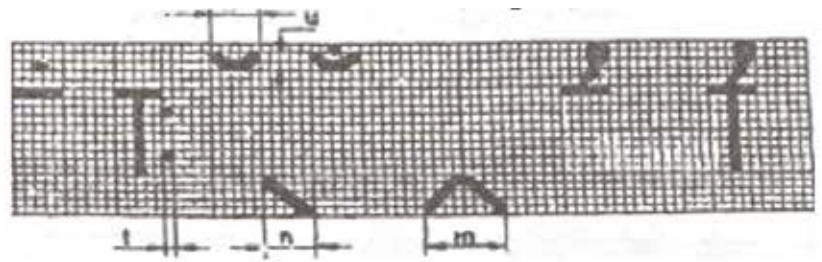


Microsoft Indic tools

Microsoft provides a number of tools that allow you to type in your native language in any MS app <https://digit.in/sep19-42>

is very important to keep in mind is - how exactly, somebody who would use their language, use the keypad to type, and typing is equivalent to what you would have written on paper. What happens is that for any user, the most natural way of thinking, is the intuitive process of writing. If they are able to think of that, it becomes easy to try to translate the same into the digital process. For example, when you learn to write to English, you are learning only the letters. Once you have learnt the letters, let's say, "apple", you need the spelling to be able to write it. Whereas when it comes to Indian languages, you don't learn writing like this. You do learn the varnamala, and actually end up realising that there are certain letters which do not form any words. No words start with those letters at all. This particular property for Indian languages is fixed. If words cannot start with that letter, they just cannot. That's it. Period. You can't even create an imaginary name or a word that starts with the letter. An example is the क (ka) varg nasal, ख (xn). You cannot find any word that starts with that nasal, and you cannot even create one. There are many such examples for every language. That is one property of the language itself, the way that you learn varnamala. Similarly you also learn barakhadi (the matras joining the consonants), and then you learn yuktakshar - when two consonants join and form either a conjunct or a composite consonant. An example is the word मुक्ति. Yuktakshars also have different behaviors, like र (ra) when it joins with a consonant, when it precedes it forms reph, when it succeeds it forms rakar. Only when have learned all of it, then we are ready to write." The word रघु is an example of reph or rafar, which are unique to the character र (ra). There are three types of rakars, which are positioned based on the shape of the base letter. These are the bottom (राष्ट्र), side (चक्र) and inside (ह्रास) rakars.

When designing a keyboard for Indic languages, the idea is to allow the knowledge of how to write, to translate



Hindi diacritics, repha and rafar

intuitively into the typing experience. Taking into account the practicality of using the language in India, allows for the creation of an intuitive typing experience overall.

LAYOUT

The Centre for Development of Advanced Computing (CDAC), which is the research and development organisation of the Ministry of Electronics and Information Technology (MeitY), has developed an Indic language smartphone keyboard for Android users. This keyboard directly translates the INSCRIPT overlay for QWERTY keyboards (think of it as a layer of stickers that could be applied on top of regular QWERTY keyboards). The INSCRIPT keyboard is a standard that was developed by the Government of India, for text input in Indian languages. The layout is made available through the Indian Language Data Centre (ILDC) platform. The keyboard is a good example of the constraints of designing a layout for Indic languages. There are thirteen narrow keys per row, and five rows (as against three rows of ten keys each). For some languages, the characters are actually bigger than the keys, and go beyond the boundary of each key. This is a keyboard layout that definitely benefits from being used in the landscape mode, but that comes at the cost of less space for reading the entered text. Each key is narrow, making it easier for users to tap on the wrong keys. Navigating through the text is achieved through left and right arrows. As can be seen here, one of the biggest challenges to designing

Indic language keyboards, is to find enough space to fit in all the keys in the vast character sets. The keyboard does support all the 22 official languages as well. For Manipuri and Santhali, there are options available to use the Meetei Mayek and the Ol Chiki script.

Google's Gboard, which replaced the Google Indic Keyboard, has support for all the 22 official languages too. Additionally, there are more obscure languages including Awadhi, Bagheli, Bagri, Dhatki, Gondri, Gujar, Khasi, Kumaoni, and Magahi. There are ten keys per row, with five rows. The consonants are arranged in sets of five. Originally, there were special keys for the matras, but now the matras appear on a top row after a consonant has been entered. For some languages, there are more than one pane of characters, which can be cycled through using a button that indicates the number of panels as well as the current panel (eg: 2/3).

Daan van Esch, Technical Program Manager at Google explains the process of designing the layout, "layout design for a new language on Gboard requires careful investigation and



The CDAC keyboard in Malayalam